

JEE MAIN REVISION MASTER: WAVE OPTICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 26 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Huygens' Principle & Interference (YDSE)	15 PYQs	Interference
High	Diffraction & Polarization (Brewster's & Malus' Law)	13 PYQs	Diffraction & Polarization

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Huygens' Principle & Interference (YDSE)	$\Delta x = \frac{y}{D}$, $\beta = \frac{\lambda D}{d}$, $I_{\text{max}} = (I_1 + I_2)$, $\text{Shift due to slab} = \frac{(\mu - 1)t}{d}$
Diffraction & Polarization (Brewster's & Malus' Law)	$a \sin \theta = n\lambda$ (minima), $\beta_0 = \frac{2\lambda D}{a}$ (central max), $I = I_0 \cos^2 \theta$ (Malus), $\tan i_p = \mu$ (Brewster)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Huygens' Principle & Interference (YDSE) (Very High Yield — 15 PYQs)

Core Principle: Wavefront geometry, Young's double slit interference, fringe width $\beta = \lambda D/d$, and fringe shift on introducing thin mica sheet.

■ **Caution / JEE Pitfall:** Fringe width β depends on wavelength ($\beta \propto \lambda$). If YDSE is immersed in water ($\mu = 4/3$), fringe width decreases by factor of $4/3$.

■ **Speed Trick:** Number of fringes shifted when a transparent slab of thickness t is placed: $N = ((\mu - 1)t / \lambda)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In Young's double slit experiment, how many maxima can be seen on a screen (including central maxima) if $d = (5\lambda / 2)$ (where λ is wavelength)?" [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "If A is incident plane wave front then refracted wave front in ($n_2 > n_1$)..." [Ans: N/A]
- [2024 Shift] Q#4 (MCQ): "In Young's double slit experiment, an electron beam is used to produce interference fringes of width β_1 . Now the electron beam is ..." [Ans: N/A]

3.2 Diffraction & Polarization (Brewster's & Malus' Law) (High Yield — 13 PYQs)

Core Principle: Fraunhofer single-slit diffraction, width of central bright maximum, Brewster's angle, and Malus' intensity law for polarized light.

■ **Caution / JEE Pitfall:** In single slit diffraction, minima occur at $a \sin \theta = n\lambda$, whereas in YDSE interference, maxima occur at $d \sin \theta = n\lambda$!

■ **Speed Trick:** At Brewster's angle of incidence i_p , the reflected ray and refracted ray are mutually PERPENDICULAR ($i_p + r = 90^\circ$).

Representative Source PYQs:

- **[2024 Shift] Q#3 (MCQ):** "Three polaroid sheets are co-axially placed as indicated in the diagram. Pass axes of the polaroids 2 and 3 make 30° and 90° ..." **[Ans: B]**
- **[2023 Shift] Q#2 (MCQ):** "An unpolarised light of intensity I is passed through two polaroids kept one after the other with their planes parallel to each other. The..." **[Ans: N/A]**
- **[2022 Shift] Q#2 (MCQ):** "In case of Fraunhofer diffraction at a single slit, the diffraction pattern on the screen is correct for which of the following statements?..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In Young's double slit experiment, how many maxima can be seen on a screen (including central maxima) if $d = (5\lambda / 2)$ (where λ is wavelength of light and d is distance between the two slits).

- (A) 5
- (B) 4
- (C) 7
- (D) 1

Correct Answer: None

Solution Analysis: For bright fringes or maxima in Young's double slit experiment, the path difference must be an integral multiple of wavelength: $d \sin \theta = n\lambda$ where $n = 0, \pm 1, \pm 2, \dots$. Since the maximum possible value of $\sin \theta$ is 1, $|n\lambda| \leq d$. Given, $d = (5\lambda / 2)$. So, $|n| \leq (d / \lambda)$ $|n| \leq (5 / 2)$ $|n| \leq 2.5$. But n must be an integer. So possible values of n are $n = -2, -1, 0, 1, 2$. That gives total maxima: 5. So the correct option is: 5 Option A.

Benchmark Problem 2: KCET 2024 (Q#3)

Three polaroid sheets are co-axially placed as indicated in the diagram. Pass axes of the polaroids 2 and 3 make 30° and 90° with pass axis of polaroid sheet 1. If I_0 is the intensity of the incident unpolarised light entering sheet 1, the intensity of the emergent light through sheet 3 is

- (A) zero
- (B) $\frac{3I_0}{32}$ **[Correct]**
- (C) $\frac{3I_0}{8}$
- (D) $\frac{3I_0}{16}$

Correct Answer: B

Solution Analysis: For unpolarised light falling on the first polaroid, the transmitted intensity becomes half. So after sheet 1, $I_1 = \frac{I_0}{2}$. Now the pass axis of sheet 2 makes an angle 30° with sheet 1. Using Malus' law, $I_2 = I_1 \cos^2 30^\circ$. Since $\cos 30^\circ = \frac{\sqrt{3}}{2} \Rightarrow \cos^2 30^\circ = \frac{3}{4}$ we get $I_2 = \frac{I_1}{2} \cdot \frac{3}{4} = \frac{3I_1}{8} = \frac{3I_0}{16}$. Now sheet 3 makes 90° with sheet 1, while sheet 2 makes 30° with sheet 1. So the angle between sheets 2 and 3 is $90^\circ - 30^\circ = 60^\circ$. Again us